# ELECTROMAGNETISM: ANSWERS

NB: this is an open-book examination. Students have access to lecture notes that contain background equations.

**Question 1. [a) = 15], [b) = 7], [c) = 10], [d) = 8]**

a) (calculated example)

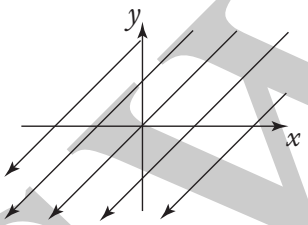
i)

$$\mathbf{E} = -\nabla\varphi = -\mathbf{e}_x - \mathbf{e}_y \quad [3]$$

ii)

$$\frac{dy}{dx} = 1 \quad y = x + c \quad [4]$$

iii)




[4]

iv)

$$\rho = \nabla \cdot \mathbf{E} = 0 \quad [4]$$

b) (calculated example). The expressions for the reflection, $\rho$, and transmission, $\tau$, coefficients of a junction between two lines with impedances $Z_1$ and $Z_2$ are (incidence from $Z_1$)

$$\rho = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad [1]$$

$$\tau = \frac{2Z_2}{Z_2 + Z_1} \quad [1]$$

We have $Z_{1,2} = \sqrt{L/C_{1,2}}$ [1] so that

$$\rho = \frac{\sqrt{L/C_2} - \sqrt{L/C_1}}{\sqrt{L/C_2} + \sqrt{L/C_1}} = \frac{1/2 - 1}{1/2 + 1} = -\frac{1}{3} \quad [2]$$

$$\tau = \frac{2\sqrt{L/C_2}}{\sqrt{L/C_2} + \sqrt{L/C_1}} = \frac{1}{1 + 1/2} = \frac{2}{3} \quad [2]$$

Answers

A different version of the paper is asking to find the same coefficients for incidence from line 2. Then

$$\rho = \frac{Z_1 - Z_2}{Z_2 + Z_1} \quad [1]$$

$$\tau = \frac{2Z_1}{Z_2 + Z_1} \quad [1]$$

We have $Z_{1,2} = \sqrt{L/C_{1,2}}$ [1] so that

$$\rho = \frac{\sqrt{L/C_1} - \sqrt{L/C_2}}{\sqrt{L/C_2} + \sqrt{L/C_1}} = \frac{1 - 1/2}{1/2 + 1} = \frac{1}{3} \quad [2]$$

$$\tau = \frac{2\sqrt{L/C_1}}{\sqrt{L/C_2} + \sqrt{L/C_1}} = \frac{1}{1 + 1/2} = \frac{4}{3} \quad [2]$$

c) (derivation, reasoning)

$$v = \frac{\omega}{k_z} = c\sqrt{1 + \left(\frac{\pi n}{k_z d}\right)^2} \quad [3]$$

We have $v \to c$ when $k_z \to \infty$ [2]. The explanation is as follows. When $k_z$ is large, the wavelength is small, and the waveguide height is very large compared to the wavelength. The interior of the waveguide then behaves like free space, in which EM waves propagate with the velocity of light [5].

d) (derivation, reasoning, partially bookwork) In the far-field region ($kr \gg 1$), we can see the following. First, terms in the brackets containing $1/r$, $1/r^2$ are small and can be ignored. Second, $E_r$ decays faster than the other two components and can be ignored. Therefore [3]

$$H_\varphi = j\frac{I_0 kd}{4\pi}\frac{e^{-jkr}}{r}\sin\theta$$
$$E_r = 0$$
$$E_\theta = j\frac{I_0 k^2 d}{4\pi\varepsilon_0\omega}\frac{e^{-jkr}}{r}\sin\theta = jZ_0\frac{I_0 kd}{4\pi}\frac{e^{-jkr}}{r}\sin\theta$$

In the near field-region ($kr \ll 1$), we can keep the largest terms in the brackets ($1/r$ or $1/r^2$) and ignore $jkr$ in the exponents. Therefore [3]

$$H_\varphi = \frac{I_0 d}{4\pi r^2}\sin\theta$$
$$E_r = -j\frac{I_0 d}{2\pi\varepsilon_0\omega r^3}\cos\theta$$
$$E_\theta = -j\frac{I_0 d}{4\pi\varepsilon_0\omega r^3}\sin\theta$$

Comparing the two sets of expressions to each other, we see that in the near-field region the phases of the field components do not vary with distance, and their magnitudes decay faster with distance than the magnitudes of the fields in the far-field region [2].

Answers

**Question 2. [a) = 15], [b) = 15]**

a) (derivation) Assume the wave propagates along the $z$-direction, and $\mathbf{E}$ is along the $y$-axis. Then the total $\mathbf{E}$ can be written as

$$\mathbf{E} = \left(Ie^{-jkz} + Re^{jkz}\right)\mathbf{e}_x \quad [2]$$

Now, find $\mathbf{H}$:

$$\mathbf{H} = -\frac{\nabla \times \mathbf{E}}{j\omega\mu_0} = -\frac{jk}{\omega\mu_0}\left(Ie^{-jkz} - Re^{jkz}\right)\mathbf{e}_y = -\frac{j}{Z_0}\left(Ie^{-jkz} - Re^{jkz}\right)\mathbf{e}_y \quad [4]$$

Assuming the boundary is at $z = 0$, the boundary condition $H_y = 0$ yields

$$I - R = 0$$

from which the wave reflection coefficient is 1 [3]. Then

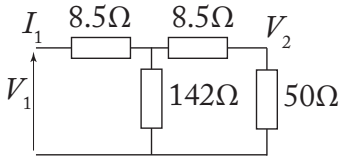
$$\mathbf{H} = \frac{2I}{Z_0}\sin(kz)\mathbf{e}_y \quad [3]$$

and

$$\mathbf{E} = 2I\cos(kz)\mathbf{e}_x \quad [3]$$

Answers

c) (calculated example) Refer to the figure below for definitions




The S-parameters for incidence from port 1 can be found as

$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \quad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1} \quad [2]$$

To find $I_1$, calculate the total impedance

$$Z_{tot} = 8.5 + \frac{142 \times 58.5}{142 + 58.5} = 49.9\ [\Omega] \quad [3]$$

Then $I_1 = V_1 / Z_{tot}$ and

$$S_{11} = \frac{1 - 50/49.9}{1 + 50/49.9} \approx 0 \quad [4]$$

Reflection is negligible. Voltage $V_2$ can be found from standard circuit analysis as

$$V_2 = (1 - \frac{8.5}{49.9})\frac{50}{58.5} V_1 = 0.71 V_1 \quad [2]$$

then

$$S_{21} = 0.71 \quad [1]$$

Because the device is reciprocal, $S_{22} = S_{11}$ and $S_{12} = S_{21}$. [2]

To determine the function of the device, note that $|S_{21}|^2 = 0.5$. Since there is no reflection, the device absorbs half of the input power. It is a 3 dB attenuator. [2]

Answers

**Question 3. [a) = 2], [b) = 8], [c) = 8], [d) = 12]**

a) (reasoning) Because the input impedance is purely imaginary, there is no loss. All input power is reflected back into the VNA, and $|S_{11}| = 1$ (or 0 dB). [2]

b) (derivation) Denote the voltage and current at the output of the network analyser by $V$ and $I$, respectively. Then $Z = V/I$. On the other hand

$$V = V^+ + V^- \quad [1]$$

and

$$I = \frac{V^+ - V^-}{Z_0} \quad [2]$$

where $V^+$ and $V^-$ are the voltages of the incident and reflected waves, respectively, and $Z_0 = 50\,\Omega$. Substituting gives

$$Z = Z_0 \frac{V^+ + V^-}{V^+ - V^-} = Z_0 \frac{1 + \frac{V^-}{V^+}}{1 - \frac{V^-}{V^+}} = Z_0 \frac{1 + S_{11}}{1 - S_{11}} \quad [5]$$

c) (calculation) The expression for the input impedance is

$$Z = Z_0 \frac{Z_L + jZ_0 \tan(kd)}{Z_0 + jZ_L \tan(kd)} \quad [1]$$

where $Z_L = j\omega L + 1/(j\omega C)$ and $k$ is the wavenumber $k = \omega/c$. We know the frequency at which Im $Z = 0$. Denote it by $f_0 = 2\pi\omega_0$. Then

$$\omega_0 L - \frac{1}{\omega_0 C} + Z_0 \tan \frac{\omega_0 d}{c} = 0 \quad [4]$$

From here

$$C = \frac{1}{\omega_0 \left( \omega_0 L + Z_0 \tan \frac{\omega_0 d}{c} \right)} = 1 \text{ nF} \quad [3]$$

d) (reasoning) The following arguments may be put forward:

1. For $d = 15$ cm, $kd \approx \tan(kd) \approx 0.31$ at 100 MHz. There are no resonances due to the cable length, and there will be none if the length is changed to either 10 or 20 cm. Therefore, when the length of the cable is changed, the behaviour of Im $Z$ will be qualitatively similar: negative values at the lower end, positive at the upper end, zero in between. [3]

2. At the lowest frequency range $kd$ is small and can be neglected in the expression

$$Z = Z_0 \frac{Z_L + jZ_0 \tan(kd)}{Z_0 + jZ_L \tan(kd)}$$

We can expect that the used cable lengths will not affect the value of the impedance appreciably. Therefore, the curve for Im $Z$ will start from approximately the same position with a different cable length. [3]

3. The frequency at which Im $Z = 0$ will change. The condition for this frequency is

$$\omega_0 L - \frac{1}{\omega_0 C} + Z_0 \tan \frac{\omega_0 d}{c} = 0$$

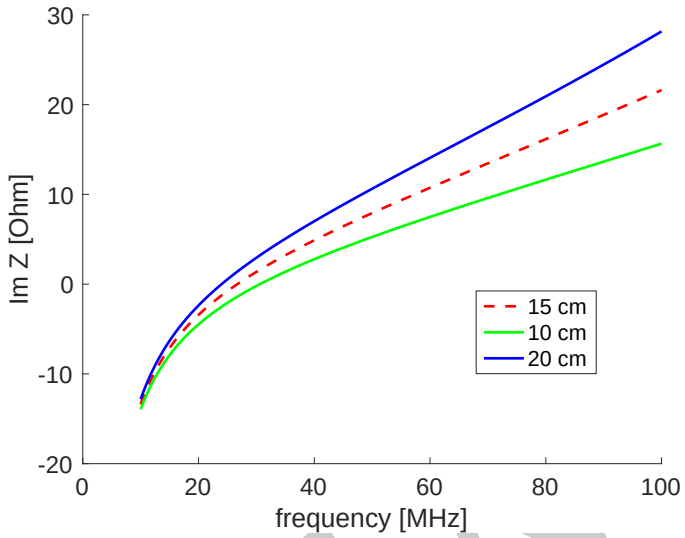
Answers

Given that $\tan(kd) \approx kd$ we can rewrite this expression as

$$\omega_0 \left( L + Z_0 \frac{d}{c} \right) - \frac{1}{\omega_0 C} = 0$$

Changing the cable length is therefore equivalent to changing the inductance. Increasing the length (to 20 cm) will therefore cause $\omega_0$ to decrease. Decreasing the length (to 10 cm) will have the opposite effect. [3]

4. Using similar logic as in 3, we can say that a longer cable will increase impedance at the high frequencies. A shorter cable will decrease the impedance. [3]




Answers